

CHReME C-DAC's HPC Resource Management Engine



Modern-day scientists and researchers rely on state-of-the-art HPC systems to solve complex and time consuming problems, greatly increasing the pace of scientific research. However, scientists require some amount of expertise in Linux and HPC systems. This article introduces the Web-based HPC portal, CHReME, which offers a user-friendly GUI and enables the easy use and management of HPC systems. The aim is to reduce the burden on scientists and to enable them to focus on their areas of research.

"Simplicity is the ultimate sophistication." - Leonardo da Vinci

CHReME, an indigenous HPC product, developed by C-DAC, the premier R&D organisation under the Ministry of Communications and IT, has bagged the IDC HPC Innovation Excellence Award. This was announced at the International Data Corporation (IDC) briefing on the sidelines of the Supercomputing Conference 2011 held at Seattle, Washington state.

An overview

High Performance Computing (HPC) is a set of algorithms, hardware and software that solve highly challenging, complex, time-consuming and advanced

computational problems. These HPC systems are the workhorses that carry out research and development activities for scientific and social advancement. HPC has permeated the fields of computational science and engineering, covering computational fluid dynamics, computational physics and chemistry, and computer science, with a special emphasis on industrial applications. According to Goldi Misra, group co-ordinator and head of the High Performance Computing and Solutions Group, C-DAC Pune, to access Linux-based HPC environments, scientists and researchers require expertise in Linux and HPC, which is something many do not have. This has